

Chapter 1: The Laws of Thermodynamics

The first law of thermodynamics states that energy cannot be created or destroyed, only converted from one form to another. The internal energy of a closed system changes by the heat added to the system minus the work done by the system.

The second law of thermodynamics states that the entropy of an isolated system never decreases. Entropy is a measure of disorder, and heat flows spontaneously from hot bodies to cold bodies, never the other way without external work.

Chapter 2: Heat Transfer

Heat transfer occurs by conduction, convection and radiation. Conduction is the transfer of heat through a solid material by molecular vibration, described by Fourier's law.

Convection is heat transfer by the movement of a fluid, and radiation is energy emitted as electromagnetic waves, described by the Stefan-Boltzmann law.